

Overview

Gaffer publishes a small, externally accessible, read-only API of generic coaching reference content: supported football formations and tactical approaches. It requires no login and returns no team, player, or account data — see the security boundary below.

BASE URLs

```
Frontend (proxied): https://gaffer-virid.vercel.app/api/v1/formations
Backend (direct):  https://gaffer-api-ynaf.onrender.com/v1/formations
```

Security & access

- No authentication required — every route here is public by design.
- GET only. There are no create, update or delete endpoints on this API.
- CORS is open (Access-Control-Allow-Origin: *) for these two routes specifically; every other Gaffer endpoint keeps its normal cookie-authenticated, allow-listed CORS policy.
- Responses are sanitized, static reference data — formation shapes and tactical style descriptions. No player, coach, team, email, or session data is ever returned.

Response envelope

Both endpoints share one envelope. Listing returns every record; adding ?id= narrows it to one, still wrapped in a single-item data array.

SHAPE

```
{
  "success": true,
  "count": <number of records returned>,
  "data": [ { ...record } ]
}
```

Interactive docs

Swagger UI documents these routes under the "Public API" tag at <https://gaffer-api-ynaf.onrender.com/api/docs> after deployment. No account, API key or token is required. Local docs: <http://localhost:3000/api/docs>.

GET /v1/formation

Returns the catalog of supported formations, or a single formation when ?id= is given.

Query parameters

Name	Type	Required	Description
id	string	No	Formation id to fetch, e.g. "4-3-3". Empty string is a 400.

Example: list all formations

REQUEST

```
GET https://gaffer-virid.vercel.app/api/v1/formation
```

RESPONSE

```
HTTP 200 OK

{
  "success": true,
  "count": 8,
  "data": [
    {
      "id": "4-3-3",
      "name": "4-3-3",
      "shape": "4-3-3",
      "description": "A balanced formation with width in attack and a three-player midfield that can dominate possession."
    }
    // ...7 more formations
  ]
}
```

Example: one formation by id

REQUEST

```
GET https://gaffer-virid.vercel.app/api/v1/formation?id=4-3-3
```

GET /v1/formation

Football formation reference data

GAFFER

PUBLIC API REFERENCE · v1.0.0

RESPONSE

HTTP 200 OK

```
{
  "success": true,
  "count": 1,
  "data": [
    {
      "id": "4-3-3",
      "name": "4-3-3",
      "shape": "4-3-3",
      "description": "A balanced formation with width in attack and a three-player midfield that can dominate possession."
    }
  ]
}
```

GET /v1/tactics

Returns the catalog of supported defensive/offensive tactical styles, or a single tactic when ?id= is given. formationId is always null — a style applies across every formation, not one specific shape.

Query parameters

Name	Type	Required	Description
id	string	No	Tactic id to fetch, e.g. "possession". Empty string is a 400.

Example: one tactic by id

REQUEST

```
GET https://gaffer-virid.vercel.app/api/v1/tactics?id=possession
```

RESPONSE

HTTP 200 OK

```
{
  "success": true,
  "count": 1,
  "data": [
    {
      "id": "possession",
      "name": "Possession",
      "category": "offensive",
      "description": "Short passing and support runs to keep the ball rather than break early.",
      "formationId": null
    }
  ]
}
```

Tactic categories

category is either "defensive" (5 styles: drop back, balanced, pressure on heavy touch, press after possession loss, constant pressure) or "offensive" (4 styles: possession, balanced, fast build up, long ball).

Error responses

Status	Cause	Body
400	?id= present but empty	{ statusCode, message, error: "Bad Request" }
404	Unknown id	{ statusCode, message, error: "Not Found" }
500	Unhandled server error	Nest's default internal error body

PublicFormation fields

Field	Type	Description
id	string	Stable identifier, e.g. "4-3-3". Mirrors FORMATION_IDS.
name	string	Display name.
shape	string	The formation shape string, e.g. "4-3-3".
description	string	One-line description of the formation.

PublicTactic fields

Field	Type	Description
id	string	Stable identifier, e.g. "possession".
name	string	Display name.
category	"defensive" "offensive"	Which side of play the style applies to.
description	string	The style's trade-off, one line.
formationId	string null	Always null — styles are formation-agnostic.